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PROCESSING APPARATUS AND WAFER PROCESSING METHOD

Abstract

A processing apparatus includes a cassette stage for mounting a cassette thereon, a chuck table to suction-hold a wafer, a processing unit to process the wafer that is held on the chuck table, a processing table that is used to perform a predetermined processing on a wafer, a robot equipped with a robot hand that has a wafer presence sensor to detect a presence or absence of a wafer and holds a wafer to unload or load the wafer from or into the cassette mounted on the cassette stage, and a controller that uses the wafer presence sensor of the robot hand to detect whether a wafer is present on the processing table.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from Japanese Patent Application No. 2024-32860 filed with the Japan Patent Office on Mar. 5, 2024, and from Japanese Patent Application No. 2024-201531 filed with the Japan Patent Office on Nov. 19, 2024, the entire contents of both of which are hereby incorporated.

BACKGROUND ART

1. Technical Field

[0002] The present invention relates to a processing apparatus and a wafer processing method.

2. Related Art

[0003] As disclosed in JP 2021-126744A and JP 2021-132180A, in a processing apparatus such as a grinding apparatus for grinding a wafer, a wafer is transferred from a cassette mounted on a cassette stage to a chuck table, where the wafer held on the chuck table is ground by a grinding stones. Subsequently, the wafer is transferred from the chuck table to a spinner table of a spinner cleaning unit, cleaned on the spinner table, and then stored in a cassette.

SUMMARY

[0004] When holding and cleaning a wafer on a spinner table, if the wafer is warped, the processing apparatus may recognize that the wafer is not held on the spinner table and halt the cleaning operation. That is, although the wafer is present on the spinner table, the processing apparatus may occasionally determine that no wafer is present on the spinner table.

[0005] If a continuous processing proceeds in this state, a next wafer is mounted on top of the wafer already on the spinner table, leading to potential separation and damage of the overlapping wafer while cleaning in a spinner cleaning unit.

[0006] Accordingly, a purpose of the present invention is to accurately detect the presence or absence of a wafer on a table such as a spinner table.

[0007] A processing apparatus according to one aspect of the present invention (the present processing apparatus) includes a cassette stage for mounting a cassette thereon, a chuck table to suction-hold a wafer, a processing unit to process the wafer that is held on the chuck table, a processing table that is used to perform a predetermined processing on a wafer, a robot equipped with a robot hand that has a wafer presence sensor to detect a presence or absence of a wafer and holds a wafer to unload or load the wafer from or into the cassette mounted on the cassette stage, and a controller that uses the wafer presence sensor of the robot hand to detect whether a wafer is present on the processing table.

[0008] In the present processing apparatus, the processing table may be the spinner table that suction-holds a wafer to perform the cleaning process on the wafer. This specific embodiment of the present processing apparatus is referred to as a first processing apparatus.

[0009] In the first processing apparatus, the controller may use the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the spinner table while relatively moving the spinner table and the robot hand positioned above the spinner table in a horizontal direction.

[0010] The first processing apparatus may further include a pressure sensor that measures a pressure value in a suction pipe connecting the spinner table and a suction source. In this case, when the spinner table suction-holds the wafer, if the measurement of the pressure sensor does not fall no more than a predetermined threshold, the controller may use the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the spinner table.

[0011] In the present processing apparatus, the processing table may be a positioning table for positioning a wafer to be held on the chuck table. This specific embodiment of the present processing apparatus is referred to as a second processing apparatus.

[0012] In the second processing apparatus, the controller may use the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the positioning table while relatively moving the positioning table and the robot hand, which is positioned above the positioning table in a horizontal direction.

[0013] The second processing apparatus may further include a pressure sensor that measures a pressure value in a suction pipe connecting the positioning table and a suction source. In this case, when the positioning table suction-holds the wafer, if the measurement of the pressure sensor does not fall no more than the predetermined threshold, the controller may use the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the positioning table.

[0014] A first wafer processing method according to one aspect of the present invention is the wafer processing method for processing the wafer using the first processing apparatus, which includes holding the wafer on the chuck table, processing the wafer with the processing unit, holding the processed wafer on the spinner table and cleaning the wafer, and detecting the presence or absence of a wafer on the spinner table using the wafer presence sensor of the robot hand at a predetermined timing before the cleaning of the wafer, where during the detection of the presence or absence of a wafer on the spinner table, if a wafer is detected to be absent from the spinner table, cleaning of the wafer is performed, while during the detection of the presence or absence of a wafer, if a wafer is detected to be present on the spinner table, a notification unit of the processing apparatus notifies an operator that a wafer is present on the spinner table.

[0015] A second wafer processing method according to another aspect of the present invention is a wafer processing method for processing the wafer using the second processing apparatus, which includes unloading a wafer from the cassette and holding the wafer on the positioning table to perform positioning of the wafer, holding the positioned wafer on the chuck table, processing the wafer with the processing unit; and detecting the presence or absence of a wafer on the positioning table using the wafer presence sensor of the robot hand at a predetermined timing before the positioning of the wafer, where during the detection of the presence or absence of a wafer on the positioning table, if a wafer is detected to be absent from the positioning table, positioning of the wafer is performed, while during the detection of the presence or absence of a wafer, if a wafer is detected to be present on the positioning table, a notification unit of the processing apparatus notifies an operator that a wafer is present on the positioning table.

[0016] In the present processing apparatus and the first and second wafer processing methods, the wafer presence sensor of the robot hand detects the presence or absence of a wafer on the processing table (for example, the spinner table or the positioning table). As a result, the presence or absence of the wafer on the processing table can clearly be detected. Consequently, for example, situations where an unsuitable wafer remains on the processing table and an overlapping wafer is mounted thereon causing damage to the wafer, can be effectively suppressed. Furthermore, the need for an operator to visually confirm the presence or absence of the wafer on the processing table is eliminated, thereby improving operational efficiency.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a perspective view illustrating a structure of a grinding apparatus.

[0018] FIG. 2 is a cross-sectional view illustrating a configuration of a spinner table.

[0019] FIG. 3 is a cross-sectional view illustrating a configuration of a positioning table.

DETAILED DESCRIPTION

[0020] In the following detailed description, for purpose of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details.

In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

[0021] A grinding apparatus **1** shown in FIG. **1** is an example of a processing apparatus. The grinding apparatus **1** grinds a wafer **100** held on a chuck table **5** using a rough grinding mechanism **30** and a finish grinding mechanism **31**.

[0022] The wafer **100** shown in FIG. **1** is an example of a workpiece and may be, for instance, a circular semiconductor wafer. On a surface **101** of the wafer **100**, which faces downward in FIG. **1**, a device (not shown) is formed. A back surface **103** of the wafer **100** serves as a processed surface subjected to grinding.

[0023] The grinding apparatus **1** includes a first apparatus base **10** and a second apparatus base **11** disposed behind the first apparatus base **10** (+Y direction side). An area above the first apparatus base **10** serves as a loading and unloading area **401** for the wafer **100**, where the wafer **100** is loaded and unloaded. A processing area **402** is positioned above the second apparatus base **11**. In a processing area **402**, the rough grinding mechanism **30** and the finish grinding mechanism **31** process the wafer **100** that is held on the chuck table **5**.

[0024] A first cassette stage **160** and a second cassette stage **162** for mounting the cassettes are provided at a front side (−Y direction side) of the first apparatus base **10**. A first cassette **161** and a second cassette **163**, each housing the wafer **100**, are mounted on the first cassette stage **160** and the second cassette stage **162**, respectively.

[0025] The first cassette **161** and the second cassette **163** are provided with multiple shelves inside, with each shelf accommodating a single wafer **100**.

[0026] The openings (not shown in the figures) of the first cassette **161** and the second cassette **163** face the +Y direction. A robot **150** is positioned on the +Y direction side of these openings.

[0027] The robot **150** is equipped with a robot hand **151** that holds the wafer **100** for unloading or loading the wafer **100** from or into the first cassette **161** and the second cassette **163**. The robot hand **151** features a U-shaped plate structure and is equipped with a suction surface **152** (see FIG. **2**) on one surface, allowing to hold the wafer **100** through suction.

[0028] The robot hand **151** is further equipped with a wafer presence sensor **153** on the suction surface **152** to detect the presence or absence of the wafer **100**, as well as a controller **158** to detect the presence or absence of the wafer **100**. In this embodiment, the wafer presence sensor **153** is used, for example, to detect whether the wafer **100** is present at a position opposite the suction surface **152**.

[0029] For example, the wafer presence sensor **153** includes a light emitter and a light receiver, and the light emitter emits light toward the predetermined position opposite the suction surface **152**, while the light receiver receives the reflected light that the light emitter emits. The controller **158** determines the presence or absence of the wafer **100** at the predetermined position based on the amount of reflected light received. The controller **158** may also serve as the controller **7** of the grinding apparatus **1**.

[0030] As shown in FIGS. **1** and **2**, the robot **150** includes a drive unit **154** that operates the robot hand **151**. The drive unit **154** controls (adjusts) the position of the robot hand **151**. Specifically, the drive unit **154** is equipped with a vertical movement mechanism **155**, a horizontal movement mechanism **156**, and an inversion mechanism **157**.

[0031] The vertical movement mechanism **155** moves the robot hand **151** vertically along the Z-axis direction. The horizontal movement mechanism **156** moves the robot hand **151** horizontally. The inversion mechanism **157** flips the robot hand **151** to position the suction surface **152** either upward or downward.

[0032] With this configuration, the robot **150** loads the processed wafers **100**, held by the robot hand **151**, to either the first cassette **161** or the second cassette **163** shown in FIG. **1**. Additionally, the robot **150** uses the robot hand **151** to unload the unprocessed wafer **100** housed in the first cassette **161** or the second cassette **163** and to mount the wafer **100** onto the positioning table **60** of

the positioning mechanism **6**.

[0033] The positioning mechanism **6** is used to position the wafer **100** (for example, detecting the center position) unloaded from the first cassette **161** or the second cassette **163** and is disposed adjacent to the robot **150**. The positioning mechanism **6** includes the positioning table **60** that holds and rotates the wafer **100** and a center position detection unit **69** equipped with a camera.

[0034] The positioning table **60** is an example of a processing table used for performing the predetermined processing (positioning of the wafer **100**) on the wafer **100**. The positioning table **60** is a table to position the wafer **100** to be held on the chuck table **5**. The positioning table **60** is communicated with a suction source **48** (see FIG. 3), thereby enabling to suction-hold the wafer **100** mounted on the positioning table **60**. The center position detection unit **69** detects the center of the wafer **100** suction-held on the positioning table **60**.

[0035] In the positioning mechanism **6**, the positioning table **60**, while suction-holding the wafer **100**, rotates at a low rotational speed, and the center position detection unit **69** captures images of multiple locations (for example, three locations) along the outer edge of the wafer **100**. The center position detection unit **69** detects the center position of the wafer **100** by performing geometric calculations based on the coordinates of the captured locations.

[0036] The positioning mechanism **6** may also include a mechanism (for example, a centering pin) for aligning the wafer **100** mounted on the positioning table **60** to the predetermined position where the center of the positioning table **60** coincides with the center of the wafer **100**.

[0037] A loading mechanism **170** is installed adjacent to the positioning mechanism **6**. The loading mechanism **170** is an example of a transport mechanism that unloads or loads the wafer **100** from or onto the chuck table **5**. The loading mechanism **170** loads the wafer **100** held by the positioning mechanism **6** onto the chuck table **5**.

[0038] The loading mechanism **170** includes a transport pad **171** that suction-holds the back surface **103** of the wafer **100**. The loading mechanism **170** uses the transport pad **171** to suction-hold the wafer **100**, which is held on the positioning table **60**, to transfer the wafer **100** onto the chuck table **5**, which is located near the positioning mechanism **6** within the processing area **402**, and to mount the wafer **100** on the holding surface **50** such that the center of the holding surface **50** aligns approximately with the center of the wafer **100**.

[0039] The chuck table **5** is a member that suction-holds the wafer **100** and has a holding surface **50**. The holding surface **50** is communicated with a suction source (not shown in the figures), enabling to suction-hold the wafer **100**. The chuck table **5** holds the wafer **100** on the holding surface **50** and, in this state, is capable of rotating around a central axis that extends in the Z-axis direction through the center of the holding surface **50**.

[0040] In this embodiment, three chuck tables **5** are arranged on the upper surface of the turntable **2** arranged on the second apparatus base **11** at equal intervals on a circle centered on a turntable **2**. At the center of the turntable **2**, a rotation shaft (not shown in the figures) for rotating the turntable **2** is arranged. The turntable **2** can rotate around the central axis, which extends in the Z-axis direction, via this rotational shaft. The rotation of the turntable **2** enables the three chuck tables **5** to revolve. As a result, the chuck tables **5** can be sequentially positioned near the positioning mechanism **6**, beneath the rough grinding mechanism **30**, and beneath the finish grinding mechanism **31**.

[0041] At the rear (+Y direction side) of the second apparatus base **11**, a first column **12** is installed upright. On the front face of the first column **12**, a rough grinding mechanism **30** for rough grinding the wafer **100** and a rough grinding feed mechanism **20** for feeding the rough grinding mechanism **30** are provided. The rough grinding mechanism **30** is an example of a processing unit that processes the wafer **100** held on the chuck table **5**, and rough grinds on the wafer **100**.

[0042] The rough grinding feed mechanism **20** moves the rough grinding mechanism **30** in a direction perpendicular to the holding surface **50**. The rough grinding feed mechanism **20** has a pair of guide rails **201** parallel to the Z-axis direction, an elevating table **203** sliding on the guide rails **201**, a ball screw **200** parallel to the guide rails **201**, a motor **202** for rotationally driving the ball

screw **200**, and a holder **204** mounted on the front face (surface) of the elevating table **203**. The holder **204** holds the rough grinding mechanism **30**.

[0043] The elevating table **203** is slidably installed on the guide rails **201**. A nut portion (not shown in the figures) is fixed to the elevating table **203**. The ball screw **200** is screwed into this nut portion. The motor **202** is connected to one end of the ball screw **200**.

[0044] In the rough grinding feed mechanism **20**, the motor **202** rotates the ball screw **200**, causing the elevating table **203** to move along the guide rails **201** in the Z-axis direction. Consequently, a holder **204** mounted on the elevating table **203**, along with the rough grinding mechanism **30** held by the holder **204**, also moves in the Z-axis direction with the elevating table **203**. In this manner, the rough grinding feed mechanism **20** feeds the rough grinding mechanism **30** in the Z-axis direction for grinding.

[0045] The rough grinding mechanism **30** has a spindle housing **301** fixed to the holder **204**, a spindle **300** rotatably held in the spindle housing **301**, a spindle motor **302** for rotationally driving the spindle **300**, a wheel mount **303** attached to a lower end of the spindle **300**, and a grinding wheel **304** detachably connected to a lower surface of the wheel mount **303**.

[0046] The spindle housing **301** is held by the holder **204** so as to be extending in the Z-axis direction. The spindle **300** extends in the Z-axis direction, perpendicular to the holding surface **50** of the chuck table **5** and is rotatably supported by the spindle housing **301**.

[0047] The spindle motor **302** is connected to the upper end side of the spindle **300**. This spindle motor **302** rotates the spindle **300** around a rotational axis extending in the Z-axis direction.

[0048] The wheel mount **303** is formed in a disk shape and is fixed to a tip (lower end) of the spindle **300**, rotating in accordance with the rotation of the spindle **300**. The wheel mount **303** supports the grinding wheel **304**.

[0049] The grinding wheel **304** is formed with an outer diameter thereof approximately equal to an outer diameter of the wheel mount **303**. The grinding wheel **304** includes an annular wheelbase **305** made of metal. On a lower surface of the wheelbase **305**, multiple approximately rectangular rough grinding stones **306** are arranged and fixed in an annular configuration around an entire circumference.

[0050] The rough grinding stones **306** are rotated because of the rotation of the spindle **300**, and the lower surfaces of the rough grinding stones **306** perform rough grinding on a back surface **103** of the wafer **100** held on the chuck table **5**. The rough grinding stones **306** contain relatively large abrasive grains.

[0051] Also, in the vicinity of the rough grinding mechanism **30**, a first measurement mechanism **25** is arranged. The first measurement mechanism **25** measures a thickness of the wafer **100** held on the chuck table **5** positioned below the rough grinding mechanism **30**.

[0052] At a rear of the second apparatus base **11**, a second column **13** is erected adjacent to the first column **12** along the X-axis direction. On a front face of the second column **13**, a finish grinding mechanism **31** for finish grinding the wafer **100** and a finish grinding feed mechanism **21** for feeding the finish grinding mechanism **31** for grinding are arranged. The finish grinding mechanism **31** is an example of a processing unit that processes the wafer **100** held on the chuck table **5**, performing finish grinding on the wafer **100**.

[0053] The finish grinding feed mechanism **21** moves the finish grinding mechanism **31** in a direction perpendicular to the holding surface **50**. The finish grinding feed mechanism **21** has the same configuration as the rough grinding feed mechanism **20** and can feed the finish grinding mechanism **31** along the Z-axis direction for grinding.

[0054] The finish grinding mechanism **31** has the same configuration as the rough grinding mechanism **30**, except that the finish grinding mechanism **31** includes a finish grinding stones **310** instead of the rough grinding stones **306**. The finish grinding stones **310** is rotated because of the rotation of the spindle **300**, and a lower surface of the finish grinding stones **310** performs finish grinding on the back surface **103** of the wafer **100** held on the chuck table **5**. The finish grinding

stones **310** is a grinding wheel containing relatively small abrasive grains.

[0055] Also, in the vicinity of the finish grinding mechanism **31**, a second measurement mechanism **26** is arranged. The second measurement mechanism **26** measures a thickness of the wafer **100** held on the chuck table **5** positioned below the finish grinding mechanism **31**.

[0056] The wafer **100** after the finish grinding is unloaded by an unloading mechanism **172**. The unloading mechanism **172** is an example of a transport mechanism that unloads or loads the wafer **100** from or onto the chuck table **5**. The unloading mechanism **172** transfers the wafer **100** held on the chuck table **5** to a spinner cleaning mechanism **8**.

[0057] The unloading mechanism **172** has a transport pad **173** that suction-holds the back surface **103** of the wafer **100**. The unloading mechanism **172** suction-holds the back surface **103** of the finish ground wafer **100** on the chuck table **5** using the transport pad **173**. The unloading mechanism **172** then unloads the wafer **100** from the chuck table **5** and transfers the wafer **100** to a spinner table **80** of the single-wafer spinner cleaning mechanism **8**.

[0058] The spinner cleaning mechanism **8** is a spinner cleaning unit that cleans the wafer **100**. The spinner cleaning mechanism **8** includes the spinner table **80** for suction-holding the wafer **100** to be cleaned and a nozzle **89** for spraying cleaning water and dry air toward the spinner table **80**.

[0059] In the spinner cleaning mechanism **8**, the spinner table **80** that holds the wafer **100** rotates, and cleaning water is sprayed onto the wafer **100**, performing spinner cleaning of the wafer **100**. Subsequently, drying air is blown onto the wafer **100** to dry the wafer **100**.

[0060] Here, the configuration of the spinner table **80** will be described. The spinner table **80** is an example of a processing table used for performing the predetermined processing (cleaning of the wafer **100**) on the wafer **100**. As shown in FIG. 2, the spinner table **80** is a circular plate-shaped table for holding the wafer **100**. The spinner table **80** has a circular plate-shaped porous member **81**, a frame **83** supporting the porous member **81**, and a table base **84** supporting the frame **83**. The porous member **81** can be communicated with the suction source **48**. The suction force from the suction source **48** is transmitted to a holding surface **82**, which is the upper surface of the porous member **81**, enabling the spinner table **80** to suction-hold the wafer **100** by the holding surface **82**.

[0061] The spinner table **80** is also rotatable by a rotation mechanism **85**. The rotation mechanism **85** is equipped with a motor **86** serving as a drive source, a shaft **87** of the motor **86**, and an encoder **88** for reading the rotational angle of the motor **86**.

[0062] The shaft **87** is connected to an underside of the spinner table **80** (table base **84**) directly below the center of the holding surface **82** and extends vertically to the holding surface **82** of the spinner table **80**. The motor **86** rotationally drives the shaft **87**, causing the spinner table **80** to rotate around the center of the holding surface **82**. Furthermore, the encoder **88** allows the rotational angle of the spinner table **80** to be recognized based on the rotational angle of the motor **86**.

[0063] The spinner table **80** is connected to a fluid circulation mechanism **40** provided in the grinding apparatus **1**. The fluid circulation mechanism **40** is a mechanism designed to supply air to the holding surface **82** of the spinner table **80** or to apply suction force to the holding surface **82**.

[0064] The fluid circulation mechanism **40** has a table internal flow path **403** provided inside the spinner table **80**, an air passage **470** communicating with the table internal flow path **403**, a rotary joint **460** connected to the shaft **87**, and an air pipe **471** communicating with the air passage **470**.

[0065] The table internal flow path **403** is arranged within the frame **83** and table base **84** of the spinner table **80** so as to contact the lower surface of the porous member **81**. The air passage **470** extends from the table internal flow path **403**, passing through the shaft **87** and the rotary joint **460**.

[0066] The air passage **470** is connected to one end of the air pipe **471** outside the rotary joint **460**. The other end of the air pipe **471** is connected to the air supply source **47**. The air supply source **47**, which is equipped with such as a compressor, is used to supply air to the holding surface **82** of the spinner table **80**.

[0067] In addition, the air pipe **471** is equipped with an air supply on-off valve **475** and an air

regulator **473** in sequence from the air supply source **47** toward the air passage **470**. The air supply on-off valve **475** switches the communication state between the air pipe **471** and the air supply source **47**. The air regulator **473**, for example, is a proportional control valve and is used to adjust the flow rate of air supplied from the air supply source **47** to the holding surface **82** by changing the orifice diameter inside thereof when the air supply on-off valve **475** is open.

[0068] Furthermore, the air pipe **471** is communicated with a suction pipe **481**. The suction pipe **481** is a pipe that communicates with the holding surface **82** of the spinner table **80** to the suction source **48**.

[0069] One end of the suction pipe **481** is connected to the air passage **470** via the air pipe **471**. The other end of the suction pipe **481** is connected to the suction source **48**. The suction source **48**, for example, is equipped with such as an ejector mechanism or a vacuum generator and is communicated with the porous member **81** of the spinner table **80** in order to apply suction force to the holding surface **82** that is an upper surface of the porous member **81**.

[0070] Furthermore, the suction pipe **481** is also equipped with a suction opening and closing valve **485** and a suction flow regulator **483** in sequence from the suction source **48** toward the air passage **470**. The suction opening and closing valve **485** switches the communication state between the suction pipe **481** and the suction source **48**. The suction flow regulator **483**, for example, is a proportional control valve and is used to adjust the suction force transmitted from the suction source **48** to the holding surface **82** of the porous member **81** by changing the orifice diameter inside thereof when the suction opening and closing valve **485** is open.

[0071] Also, it should be noted that the air regulator **473** and the suction flow regulator **483** may also be needle valves or gate valves that allow manual adjustment of the orifice diameter.

[0072] Additionally, the suction pipe **481** is equipped with a pressure sensor **487**. The pressure sensor **487** detects the pressure value in the suction pipe **481**. In this embodiment, for example, when the suction opening and closing valve **485** is opened to communicate the holding surface **82** with the suction source **48**, if the pressure sensor **487** measures a pressure value in the suction pipe **481** no more than a predetermined threshold, it is determined that the wafer **100** is mounted on the holding surface **82** of the spinner table **80** and that the wafer **100** is being held by the holding surface **82**.

[0073] The wafer **100**, cleaned by the spinner cleaning mechanism **8** having the spinner table **80** with the above-described configuration, is loaded into the first cassette **161** or the second cassette **163** (for example, the cassette from which the wafer **100** was unloaded) by the robot **150** shown in FIG. **1**.

[0074] As shown in FIG. **1**, the grinding apparatus **1** has a housing **15** that covers the first apparatus base **10** and the second apparatus base **11**. A touch panel **9** is installed on a side of the housing **15**. The touch panel **9** displays various types of information, such as conditions for processing with respect to the grinding apparatus **1**. The touch panel **9** is also used for setting various parameters. Accordingly, the touch panel **9** functions as an input member for inputting information and as a display member (notification member) for displaying (notifying) information.

[0075] The grinding apparatus **1** also includes the controller **7** therein for controlling the grinding apparatus **1**. The controller **7** is equipped with such as CPU, which performs computation in accordance with a control program, and storage media such as memory. The controller **7** controls the above-described components of the grinding apparatus **1** to perform grinding of the wafer **100**.

[0076] The wafer processing method for processing the wafer **100** using the grinding apparatus **1** that is controlled by the controller **7** will be described below.

[Positioning Process]

[0077] Firstly, a positioning process is performed. In this process, the controller **7** controls the robot **150** shown in FIG. **1** to unload an unprocessed wafer **100** from the first cassette **161** or the second cassette **163** to be held on the positioning table **60** of the positioning mechanism **6**, thereby performing positioning of the wafer **100**.

[Holding Process]

[0078] After the positioning process, a holding process is performed. In this process, the wafer **100**, which has undergone the positioning step, is held on the chuck table **5**. Specifically, the controller **7** controls the loading mechanism **170** to hold the wafer **100** on the positioning table **60** and mounts the wafer **100** with the back surface **103** facing upwards on the holding surface **50** of the chuck table **5** arranged in the vicinity of the positioning mechanism **6**. Afterward, the controller **7** communicates the holding surface **50** with the suction source (not shown in the figures). Accordingly, the holding surface **50** suction-holds the wafer **100**.

[Rough Grinding Process]

[0079] After the holding process, a rough grinding process is performed. The rough grinding process is an example of a processing process in which the wafer **100** is processed by a processing unit. In this process, the wafer **100**, which is held on the chuck table **5**, is roughly ground by the rough grinding mechanism **30** which functions as the processing unit. Specifically, after the holding process, the controller **7** rotates the turntable **2** to position the chuck table **5** holding the wafer **100** beneath the rough grinding mechanism **30**. Furthermore, the controller **7** rotates the rough grinding stones **306** of the rough grinding mechanism **30** while simultaneously rotating the chuck table **5** that holds the wafer **100**. Thereafter, the controller **7** uses the rough grinding feed mechanism **20** to lower the rough grinding mechanism **30** and to perform the rough grinding of the wafer **100**. During this process, the controller **7** uses the first measurement mechanism **25** to measure the thickness of the wafer **100**, which is being ground, and continues rough grinding until the thickness of the wafer **100** reaches the predetermined rough grinding thickness.

[Finish Grinding Process]

[0080] After the rough grinding process, a finish grinding process is performed. This finish grinding process is another example of a processing process, in which the wafer **100** is processed by a processing unit. In this process, the finish grinding mechanism **31**, which functions as the processing unit, performs the finish grinding on the wafer **100** held on the chuck table **5**. Specifically, after the rough grinding process, the controller **7** rotates the turntable **2** to position the chuck table **5** holding the rough-ground wafer **100** beneath the finish grinding mechanism **31**. Furthermore, the controller **7** rotates the finish grinding stones **310** of the finish grinding mechanism **31** while simultaneously rotating the chuck table **5** holding the wafer **100**. Afterward, the controller **7** uses the finish grinding feed mechanism **21** to lower the finish grinding mechanism **31** to perform the finish grinding of the wafer **100**. During this process, the controller **7** uses the second measurement mechanism **26** to measure the thickness of the wafer **100**, which is being ground, and continues the finish grinding process until the thickness reaches a predetermined finish grinding thickness.

[Cleaning Process]

[0081] After the finish grinding process, a cleaning process is performed. This cleaning process is a process to clean the processed wafer **100** that is held on the spinner table **80**. In this process, the controller **7** controls the turntable **2** to rotate, causing the chuck table **5** holding the finished wafer **100** to position near the spinner cleaning mechanism **8** (near the positioning mechanism **6**). Then, the controller **7** controls the loading mechanism **170** to hold the wafer **100** on the chuck table **5** and place the wafer **100** onto the holding surface **82** of the spinner table **80** of the spinner cleaning mechanism **8**. Furthermore, the controller **7** opens the suction opening and closing valve **485** shown in FIG. **2**, causing the holding surface **82** to communicate with the suction source **48**. Accordingly, the holding surface **82** suction-holds the wafer **100**.

[0082] Next, the controller **7** controls the rotation mechanism **85** to rotate the spinner table **80** while simultaneously injecting cleaning water from the nozzle **89** onto the wafer **100**, thereby performing spinner cleaning of the wafer **100**. Additionally, the controller **7** blows dry air from the nozzle **89** onto the wafer **100**, thereby drying the wafer **100**.

[0083] Afterward, the controller **7** controls the robot **150** shown in FIG. **1** to use the robot hand **151**

to hold the wafer **100** on the spinner table **80** and to house the wafer **100** in the first cassette **161** or the second cassette **163** (the cassette from which the wafer **100** was unloaded).

[0084] In this embodiment, for example, the above-described holding process, rough grinding process, finish grinding process, and cleaning process are consecutively carried out on multiple wafers **100** (continuous processing).

[0085] Additionally, in the cleaning process, when the suction opening and closing valve **485** is opened to communicate the holding surface **82** with the suction source **48** for suction-holding the wafer **100** on the holding surface **82** of the spinner cleaning mechanism **8**, for some reasons, the controller **7** may determine that the wafer **100** is not mounted on the holding surface **82**.

[0086] That is, if the wafer **100** is not mounted on the holding surface **82**, a large amount of air enters the suction pipe **481** from the holding surface **82**, and consequently, the pressure value in the suction pipe **481**, as measured by the pressure sensor **487** (the negative pressure value in the holding surface **82**), does not fall no more than the predetermined threshold.

[0087] Therefore, if the measurement of the pressure sensor **487** does not fall no more than the predetermined threshold even after the holding surface **82** is communicated with the suction source **48**, the controller **7** can determine that the wafer **100** is not mounted on the holding surface **82** of the spinner table **80**. In this case, the controller **7** can, for example, temporarily halt the continuous processing.

[0088] As shown in FIG. 2, there may be a case where the wafer **100** mounted on the holding surface **82** of the spinner table **80** is damaged and a part of the wafer **100** is missing. In such a case, an exposed portion **821**, which is a part of the holding surface **82** of the spinner table **80** not covered by the wafer **100**, may occur. Also, if the wafer **100** mounted on the holding surface **82** has a large amount of warpage, it becomes difficult for the holding surface **82** to properly hold the entire surface of the wafer **100**, and as a result, the exposed portion **821** may similarly occur on the holding surface **82**.

[0089] Hence, similar to the case where the wafer **100** is not mounted on the holding surface **82**, when a non-confirming wafer **100**, which is difficult to be properly held by the holding surface **82**, is mounted on the holding surface **82**, a large amount of air enters the suction pipe **481** from the holding surface **82** when the suction opening and closing valve **485** is opened to communicate the holding surface **82** with the suction source **48**. Accordingly, the suction pipe **481** measured by the pressure sensor **487** does not fall no more than the predetermined threshold, and as a result, continuous processing may be temporarily halted.

[0090] Furthermore, the non-confirming wafer **100** includes, for example, the wafer **100** with a missing part, the wafer **100** with warped outer edges, or the wafer **100** with cracks.

[0091] Accordingly, using only the measurement of the pressure sensor **487**, it becomes difficult to distinguish between cases where the wafer **100** is not mounted on the holding surface **82** and cases where a non-confirming wafer **100** is mounted on the holding surface **82**.

[0092] Therefore, in this embodiment, when the measurement of the pressure sensor **487** does not fall no more than the predetermined threshold and continuous processing is temporarily halted, the controller **7** performs a wafer detection process. In this process, the controller **7** controls to use the wafer presence sensor **153** of the robot hand **151** to detect whether the wafer **100** is present on the spinner table **80** before the cleaning process, specifically before transferring the wafer **100** to the spinner table **80** after the finish grinding, that is, before mounting the wafer **100** on the spinner table **80**.

[0093] In particular, when the measurement of the pressure sensor **487** does not fall no more than the predetermined threshold, the controller **7**, as shown in FIG. 2, after temporarily halting the continuous processing, controls the drive unit **154** of the robot **150** to position the suction surface **152** of the robot hand **151** above the holding surface **82** of the spinner table **80**. Then, the controller **7** uses the wafer presence sensor **153** on the suction surface **152** of the robot hand **151** to detect the presence or absence of the wafer **100** on the holding surface **82**.

[0094] At this time, for example, the controller 7 uses the wafer presence sensor 153 of the robot hand 151 to detect whether the wafer 100 is present on the spinner table 80 while relatively moving the spinner table 80 and the robot hand 151 positioned above the spinner table 80 in a horizontal direction.

[0095] For example, the controller 7 controls the rotation mechanism 85 of the spinner cleaning mechanism 8 to rotate the spinner table 80 at a low rotational speed while using the horizontal movement mechanism 156 of the drive unit 154 of the robot 150 to move the robot hand 151 horizontally. In this way, the wafer presence sensor 153 can detect the presence or absence of the wafer 100 across the entire surface of the holding surface 82.

[0096] Then, if the controller 7 determines, based on the wafer presence sensor 153, that there is no wafer 100 anywhere on the entire surface of the holding surface 82, the controller 7 concludes that the wafer 100 is not mounted on the holding surface 82 of the spinner table 80. In this case, the controller 7 can, for example, resume the continuous processing of the wafer 100 that had been temporarily halted. In other words, when the wafer 100 is detected to be absent from the spinner table 80 in the wafer detection process, the controller 7 resumes continuous processing and performs the cleaning process on the wafer 100 as using the spinner table 80 of the spinner cleaning mechanism 8 to hold the finish-ground wafer 100.

[0097] On the other hand, if the controller 7 determines, based on the wafer presence sensor 153, that there is the wafer 100 on an entire or a part of the holding surface 82, the controller 7 concludes that the non-confirming wafer 100, which is difficult to be held on the holding surface 82 properly, is mounted on the holding surface 82. In this case, the controller 7, for example, maintains the continuous processing of the wafer 100 halted and performs a notification process. In this process, the controller 7 uses the touch panel 9, which serves as a notification unit, to notify the operator that the un-confirming wafer 100 is mounted on the holding surface 82. In other words, if the controller 7 detects the presence of the wafer 100 on the spinner table 80 during the wafer detection process, the controller 7 performs the notification process via the touch panel 9 functioning as the notification unit to inform the operator that the wafer 100 is mounted on the spinner table 80.

[0098] It should be noted that, when the controller 7 determines that the wafer 100 is mounted on the entire surface of the holding surface 82 but if the wafer 100 is warped into either a convex shape or a concave shape in the middle, there may be a case where the measurement of the pressure sensor 487 does not fall no more than the predetermined threshold.

[0099] As described above, in this embodiment, when the measurement of the pressure sensor 487 does not fall no more than the predetermined threshold and continuous processing is temporarily halted, the wafer presence sensor 153 of the robot hand 151 detects the presence or absence of the wafer 100 on the holding surface 82. This enables clear detection of the presence or absence of the wafer 100 on the holding surface 82 of the spinner table 80.

[0100] Accordingly, for example, it becomes possible to effectively suppress situations where the non-confirming wafer 100 remains mounted on the holding surface 82 and continuous processing is resumed to superpose the next wafer 100 on the non-confirming wafer 100, causing damage to the wafers 100.

[0101] Additionally, there is no need for the operator to visually detect the presence or absence of the wafer 100 on the holding surface 82. This configuration improves work efficiency.

[0102] In this embodiment, the controller 7 performs the wafer detection process when the measurement of the pressure sensor 487 of the spinner cleaning mechanism 8 does not fall no more than the predetermined threshold and the continuous processing of the wafer 100, including the holding process, the processing process (the rough grinding process and the finish grinding process), and the cleaning process, is temporarily halted. Then, the controller 7 either resumes the wafer processing process to perform the cleaning process when no wafer 100 is detected on the spinner table 80 or performs the notification process when the wafer 100 is detected on the spinner

table **80**.

[0103] In this regard, the controller **7** can perform the wafer detection process at the predetermined timing set by the operator before the cleaning process. For example, the controller **7** may perform the wafer detection process when the continuous processing is temporarily halted due to the measurement of the pressure sensor **487** not falling no more than the predetermined threshold as described above, or may always or periodically temporarily halt the continuous processing and perform the wafer detection process, when performing the cleaning process on the wafer **100**, before mounting the wafer **100** on the spinner table **80**.

[0104] Additionally, in this embodiment, it is also possible to detect the presence of the wafer **100** on the positioning table **60** of the positioning mechanism **6** shown in FIG. **1**.

[0105] Here, a configuration of the positioning table **60** will be described. As shown in FIG. **3**, the positioning table **60** is a circular plate-like table for holding the wafer **100**. The positioning table **60** has a circular plate-like table plate **61** and a table base **64** that supports the table plate **61**. An upper surface of the table plate **61** is a holding surface **62** to suction-hold the wafer **100**. The holding surface **62** has a linear suction groove **63** passing through a center thereof (see FIG. **1**). The suction force from the suction source **48** is transmitted to the suction groove **63**, allowing the positioning table **60** to suction-hold the wafer **100** on holding surface **62** of the table plate **61**.

[0106] Additionally, the positioning table **60** is rotatable by a rotation mechanism **65**. The rotation mechanism **65** has a configuration similar to the rotation mechanism **85** shown in FIG. **2** and includes a motor **66** as a driving source, a shaft **67** of the motor **66**, and an encoder **68** that reads the rotational angle of the motor.

[0107] The shaft **67** is connected directly under the center of the holding surface **62** on the lower surface of the positioning table **60** (the table base **64**) and extends vertically with respect to the holding surface **62** of the positioning table **60**. As the motor **66** rotates the shaft **67**, the positioning table **60** rotates around the center of the holding surface **62**. Also, the encoder **68** can recognize the rotation angle of the positioning table **60** based on the rotational angle of the motor **66**.

[0108] In addition, the positioning table **60** is connected to a fluid circulation mechanism **40a** provided in the grinding apparatus **1**. The fluid circulation mechanism **40a** is a mechanism that supplies air to the holding surface **62** of the positioning table **60** or applies suction force to the holding surface **62**.

[0109] Since the fluid circulation mechanism **40a** has a configuration similar to the fluid circulation mechanism **40** shown in FIG. **2**, a detailed description thereof is omitted. In this fluid circulation mechanism **40a**, a table internal flow path **403** provided within the positioning table **60** is provided in the table plate **61** and the table base **64** of the positioning table **60**, and an upper end of the table internal flow path **403** is connected to the suction groove **63** formed on the holding surface **62** of the table plate **61**. Other components of the fluid circulation mechanism **40a** are similar to those of the fluid circulation mechanism **40**.

[0110] In other words, in the fluid circulation mechanism **40a**, an air passage **470** extends from the table internal flow path **403** through the shaft **67** and rotary joint **460**. The air supply source **47** or the suction source **48** is communicated with the suction groove **63** of the holding surface **62** through the air passage **470**, the air pipe **471**, and the suction pipe **481**, supplying air to the holding surface **62** or applying suction force to the holding surface **62**.

[0111] The pressure sensor **487** is provided in the suction pipe **481** in this configuration as well. Furthermore, when the suction opening and closing valve **485** is opened to communicate the suction groove **63** on the holding surface **62** with the suction source **48**, if the pressure value in the suction pipe **481** measured by the pressure sensor **487** provided in the suction pipe **481** becomes no more than the predetermined threshold, it is determined that the wafer **100** is mounted on the holding surface **62** of the positioning table **60** and is held by the holding surface **62**.

[0112] In this configuration, during the above-described holding process, when the suction opening and closing valve **485** is opened to communicate the suction groove **63** on the holding surface **62**

with the suction source **48** in order to suction-hold the wafer **100** by the holding surface **62** of the positioning mechanism **6**, there is a possibility that the wafer **100** may not be mounted on the holding surface **62** for some reasons. In this case, a large amount of air may enter the suction pipe **481** from the suction groove **63**, so the pressure value in the suction pipe **481** (negative pressure value in the suction groove **63**) measured by the pressure sensor **487** may not fall no more than the predetermined threshold.

[0113] Therefore, if the measurement of the pressure sensor **487** does not fall no more than the predetermined threshold even if the suction groove **63** is communicated with the suction source **48**, the controller **7** can determine that the wafer **100** is not mounted on the holding surface **62** of the positioning table **60**. In this case, for example, the controller **7** may temporarily halt the continuous processing.

[0114] As with the case of the spinner table **80** described above, if the wafer **100** mounted on the holding surface **62** of the positioning table **60** is damaged and a part of the wafer **100** is missing, an exposed portion **621** on the holding surface **62** (suction groove **63**), which is not covered by the wafer **100**, occurs. Similarly, if the wafer **100** mounted on the holding surface **62** has a large amount of warping, the exposed portion **621** on the holding surface **62** also occurs.

[0115] Thus, when the non-confirming wafer **100**, which is difficult to hold properly on the holding surface **62**, is mounted on the holding surface **62**, similar to when no wafer **100** is mounted, the pressure value in the suction pipe **481** measured by the pressure sensor **487** does not fall no more than the predetermined threshold, and continuous processing may be temporarily halted.

[0116] In this embodiment, when the measurement of the pressure sensor **487** does not fall no more than the predetermined threshold and continuous processing is temporarily halted, the controller **7** performs the wafer detection process. In this process, the controller **7** controls to use the wafer presence sensor **153** of the robot hand **151** to detect whether the wafer **100** is present on the positioning table **60** before the positioning process, specifically, before transporting the wafer **100** housed in the first cassette **161** or the second cassette **163** to the positioning table **60**, that is, before mounting the wafer **100** on the positioning table **60**.

[0117] Specifically, when the measurement of the pressure sensor **487** does not fall no more than the predetermined threshold, the controller **7**, after temporarily halting continuous processing, controls the drive unit **154** of the robot **150** to position the suction surface **152** of the robot hand **151** above the holding surface **62** of the positioning table **60**. as shown in FIG. 3. Then, the controller **7** uses the wafer presence sensor **153** provided on the suction surface **152** of the robot hand **151** to detect the presence of the wafer **100** on the holding surface **62**.

[0118] At this time, for example, the controller **7** uses the wafer presence sensor **153** of the robot hand **151** to detect whether the wafer **100** is present on the positioning table **60**, while relatively moving the positioning table **60** and the robot hand **151** positioned above the positioning table **60** in a horizontal direction.

[0119] For example, the controller **7** controls the rotation mechanism **65** of the positioning mechanism **6** to rotate the positioning table **60** at a low rotational speed while using the horizontal movement mechanism **156** of the drive unit **154** of the robot **150** to move the robot hand **151** horizontally. This allows the wafer presence sensor **153** to detect the presence or absence of the wafer **100** across the entire surface of the holding surface **62**.

[0120] Then, if the controller **7** determines that there is no wafer **100** anywhere on the entire surface of the holding surface **62**, the controller **7** determines that the wafer **100** is not placed on the holding surface **62** of the positioning table **60**. In this case, for example, the controller **7** resumes the temporarily halted continuous processing of the wafer **100**. That is, if the wafer **100** is detected to be absent from the positioning table **60** during the wafer detection process, the controller **7** resumes continuous processing, and the wafer **100** is unloaded from the first cassette **161** or second cassette **163** to be mounted on the positioning table **60**, performing the positioning process for that wafer **100**.

[0121] On the other hand, if the controller 7 determines that there is the wafer 100 on the entire or a part of the holding surface 62, the controller 7 concludes that the non-confirming wafer 100, which is difficult to hold properly on the holding surface 62, is mounted on the holding surface 62. In this case, for example, the controller 7 performs the notification process while maintaining the suspension of continuous processing for the wafer 100. In this process, the controller 7 uses the touch panel 9, as the notification unit for sending information to the operator, to notify the operator that the non-confirming wafer 100 is mounted on the holding surface 62. That is, when the wafer 100 is detected to be present on the positioning table 60 during the wafer detection process, the controller 7 notifies the operator using the touch panel 9 that the wafer 100 is on the positioning table 60.

[0122] As such, with regard to the suction holding of the wafer 100 by the positioning table 60, when the continuous processing is temporarily halted due to the measurement of the pressure sensor 487 not falling no more than the predetermined threshold, the wafer presence sensor 153 of the robot hand 151 can detect the presence or absence of the wafer 100 on the holding surface 62; and therefore, it is possible to clearly detect the presence or absence of the wafer 100 on the holding surface 62 of the positioning table 60.

[0123] Accordingly, for example, this configuration can effectively suppress the damage of the wafer 100 caused when the continuous processing is resumed while the non-confirming wafer 100 remains placed on the holding surface 62 and an overlapping wafer 100 is superposed thereon.

[0124] Furthermore, it is not necessarily for the operator to visually check the presence or absence of the wafer 100 on the holding surface 62. This configuration improves work efficiency.

[0125] In the above embodiment, the controller 7 performs the wafer detection process when the measurement of the pressure sensor 487 of the positioning mechanism 6 does not fall no more than the predetermined threshold and the continuous processing of the wafer 100, which includes the holding process, the processing process (the rough grinding step and the finish grinding step), and the cleaning process, is temporarily halted. Then, the controller 7 resumes the wafer processing process and performs the positioning process when there is no wafer 100 on the positioning table 60, while performing the notification process when there is the wafer 100 on the positioning table 60.

[0126] In this regard, the controller 7 can perform the wafer detection process at the predetermined timing set by the operator before the positioning process. For example, the controller 7 may perform the wafer detection process when the continuous processing is temporarily halted due to the measurement of the pressure sensor 487 not falling no more than the predetermined threshold as described above, or may always or periodically temporarily halt the continuous processing and perform the wafer detection process, when performing the positioning process on the wafer 100, before mounting the wafer 100 on the positioning table 60.

[0127] In this embodiment, when detecting the presence or absence of the wafer 100 on the holding surface 82 (the holding surface 62) of the spinner table 80 (the positioning table 60) using the wafer presence sensor 153 of the robot hand 151, the spinner table 80 (the positioning table 60) and the robot hand 151 that is positioned above the spinner table 80 are relatively moved in a horizontal direction.

[0128] In this regard, it is also possible to detect the presence or absence of the wafer 100 by arranging the wafer presence sensor 153 of the robot hand 151 above the spinner table 80 (the positioning table 60) without performing such horizontal movement. However, by performing the above-described horizontal movement, the presence or absence of the wafer 100 can be more effectively detected.

[0129] Furthermore, regarding such horizontal movement, both the rotation of the spinner table 80 (the positioning table 60) and the horizontal movement of the robot hand 151 may be performed, or only one of the rotation of the spinner table 80 or the horizontal movement of the robot hand 151 may be performed.

[0130] Also, in this embodiment, the loading mechanism **170** and the unloading mechanism **172** are shown as a transport mechanism for unloading or loading the wafer **100** from or onto the chuck table **5**. In this regard, the grinding apparatus **1** may not necessarily include the loading mechanism **170** and the unloading mechanism **172**. In this case, for example, the robot **150** may be configured to transfer the wafer **100** to and from the chuck table **5**. That is, in this case, the robot **150** may be configured to use the robot hand **151** to suction-hold the wafer **100**, which is held on the positioning table **60**, so as to be mounted on the holding surface **50** of the chuck table **5**. Also, the robot **150** may be configured to use the robot hand **151** to suction-hold the finish ground wafer **100**, which is held on the chuck table **5**, so as to be mounted on the spinner table **80** of the spinner cleaning mechanism **8**.

[0131] Furthermore, in this embodiment, the grinding apparatus **1** is illustrated as an example of the processing apparatus. In this regard, the processing apparatus in this embodiment may include other processing apparatus such as a polishing apparatus for polishing the wafer **100** and a cutting apparatus for dividing the wafer **100** into chips as long as the apparatus includes a table for suction-holding the wafer **100** and the robot **150**.

[0132] The foregoing detailed description has been presented for the purposes of illustration and description. Many modifications and variations are possible in light of the above teaching. It is not intended to be exhaustive or to limit the subject matter described herein to the precise form disclosed. Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims appended hereto.

Claims

1. A processing apparatus, comprising: a cassette stage for mounting a cassette thereon; a chuck table to suction-hold a wafer; a processing unit to process the wafer that is held on the chuck table; a processing table that is used to perform a predetermined processing on a wafer; a robot equipped with a robot hand that has a wafer presence sensor to detect a presence or absence of a wafer and holds a wafer to unload or load the wafer from or into the cassette mounted on the cassette stage; and a controller that uses the wafer presence sensor of the robot hand to detect whether a wafer is present on the processing table.
2. The processing apparatus according to claim 1, wherein the processing table is a spinner table that suction-holds a wafer to perform a cleaning process on the wafer.
3. The processing apparatus according to claim 2, wherein the controller uses the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the spinner table while relatively moving the spinner table and the robot hand, which is positioned above the spinner table, in a horizontal direction.
4. The processing apparatus according to claim 2, further comprising a pressure sensor that measures a pressure value in a suction pipe connecting the spinner table and a suction source, wherein the controller uses the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the spinner table if the measurement of the pressure sensor does not fall no more than a predetermined threshold when the spinner table suction-holds a wafer.
5. The processing apparatus according to claim 1, wherein the processing table is a positioning table for positioning a wafer to be held on the chuck table.
6. The processing apparatus according to claim 5, wherein the controller uses the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the positioning table while relatively moving the positioning table and the robot hand, which is positioned above the positioning table in a horizontal direction.

7. The processing apparatus according to claim 5, further comprising a pressure sensor that measures a pressure value in a suction pipe connecting the positioning table and a suction source, wherein the controller uses the wafer presence sensor of the robot hand to detect the presence or absence of a wafer on the positioning table if the measurement of the pressure sensor does not fall no more than a predetermined threshold when the positioning table suction-holds a wafer.

8. A wafer processing method using the processing apparatus according to claim 2, comprising: holding a wafer on the chuck table; processing the wafer with the processing unit; holding the processed wafer on the spinner table and cleaning the wafer; and detecting the presence or absence of a wafer on the spinner table using the wafer presence sensor of the robot hand at a predetermined timing before the cleaning of the wafer, wherein during the detection of the presence or absence of a wafer on the spinner table, if a wafer is detected to be absent from the spinner table, cleaning of the wafer is performed, while during the detection of the presence or absence of a wafer, if a wafer is detected to be present on the spinner table, a notification unit of the processing apparatus notifies an operator that a wafer is present on the spinner table.

9. A wafer processing method using the processing apparatus according to claim 5, comprising: unloading a wafer from the cassette and holding the wafer on the positioning table to perform positioning of the wafer; holding the positioned wafer on the chuck table; processing the wafer with the processing unit; and detecting the presence or absence of a wafer on the positioning table using the wafer presence sensor of the robot hand at a predetermined timing before the positioning of the wafer, wherein during the detection of the presence or absence of a wafer on the positioning table, if a wafer is detected to be absent from the positioning table, positioning of the wafer is performed, while during the detection of the presence or absence of a wafer, if a wafer is detected to be present on the positioning table, a notification unit of the processing apparatus notifies an operator that a wafer is present on the positioning table.
